Module 4
Energy Systems

This module discusses the circulatory system, energy systems, measurements of intensity, and the physiological effects on the body due to intense exercise.

❖ OBJECTIVES

Upon completion of this module, you should have an understanding of the following areas:

- Energy system terminology
- Circulatory system
- Function of lungs
- Role of cells in energy production
- Cori cycle
- Three energy systems
- Blood glucose regulation
- Physiological response to aerobic training
- Effect of hypoxic training
- Physiological effects of intense exercise
- Glycogen depletion

❖ TERMINOLOGY

Anaerobic: This term means *without oxygen*. This is a slight misnomer, as being in an anaerobic state typically has more to do with a lack of utilization of oxygen rather than an absence of oxygen.

Adenosine Triphosphate (ATP): A molecule that is responsible for storing and releasing energy in the body. Biologists have coined ATP the "currency of life."

ATP is generated in the mitochondria of cells and can be produced both aerobically and anaerobically.

The body stores only a small amount of ATP at any given time. Therefore, ATP must be continually regenerated on a cellular level.

When ATP is broken down for energy, one of the phosphates breaks away from the molecule. Since there are only two phosphates, the molecule is now adenosine diphosphate (ADP). Through a process called *phosphorylation*, a phosphate is added to the ADP molecule to create ATP. Phosphorylation can occur aerobically or anaerobically.

Mitochondria: An organelle that is the power plant of cells; converts potential energy from food molecules into ATP

Glucose: A simple sugar in the blood

Glycogen: Main form of carbohydrate storage in the body (muscles/liver) and can be converted to glucose

Glycolysis: Metabolism of glucose.

Pyruvate: An acid that is the result of glycolysis.

Krebs Cycle: A cycle of complex chemical reactions that cells go through in the presence of oxygen to create energy (cellular respiration). Also referred to as the *citric acid cycle*.

Lactate: An enzyme found in body tissue and a byproduct of pyruvate. During high-intensity exertion when oxygen supply is limited, pyruvate produces lactate, which allows for the breakdown of glucose for energy. At lower intensities, lactate is used as fuel by the mitochondria. Lactate also helps to buffer cellular acidosis *(refer to "Muscle Burn" in previous module for more information on buffering cellular acidosis).* Lactate is produced at all times.

Lactate Threshold: Representative of the level at which blood lactate accumulates in the bloodstream. This occurs when lactate production exceeds lactate clearing.

Maximum Lactate Steady State (MLSS): The highest intensity level at which blood lactate concentrations are maintained at a steady-state (equilibrium) level during exercise bouts of approximately 60 minutes.

Onset of Blood Lactate Accumulation (OBLA): Represented when blood lactate levels reach 4 millimoles per liter (mmol/l) during exercise bouts (125). Since this can be ascertained only by using a blood lactate analyzer, this assessment will not be used in this certification.

Acetyl-CoA: A molecule that is the result of oxidation of fatty acid, amino acids and pyruvate. Acetyl-CoA is broken down and used for energy production in the Krebs cycle.

Creatine Phosphate (CP): A high-energy phosphate compound that aids in the resynthesis or regeneration of ATP.

Ventilatory Threshold (VT): The point at which the ventilation (breathing) rate increases faster than the workload. Until the VT is reached, the workload and respiration rate increase linearly.

❖ CIRCULATORY SYSTEM

There are two main circulatory systems within the body:

1. **Pulmonary:** This system involves the heart and the lungs. Deoxygenated blood leaves the heart and goes to the lungs, where it is reoxygenated and returned to the heart.

2. **Systemic:** This system relates to circulation throughout the body with the exception of the lungs. Oxygenated blood leaves the heart via arteries to supply the body, and the deoxygenated blood comes back to the heart via the veins.

Muscles rely on a steady oxygen supply to perform aerobic activity. Oxygen is transported to and from muscles via blood vessels. There are three primary classifications of blood vessels in the body:

1. **Arteries:** Carry oxygenated blood away from the heart to the rest of the body.

2. **Veins:** Carry deoxygenated (without oxygen) blood to the heart.

3. **Capillaries:** The smallest blood vessels in the body. They increase in density based on how much blood supply is needed in a particular area. They are found in the tissues of the human body and transport blood from the arteries to the veins. Because of the thinness of the walls of capillaries, O2, CO2, waste products, and nutrients can pass through the walls. The process through which this occurs is called *diffusion* (301).

In addition to providing core muscle stability, the *diaphragm* plays a large role in the circulatory system. This sheetlike muscle contracts and relaxes to facilitate breathing. When the diaphragm is contracting, oxygen is pulled into the lungs, and when the diaphragm is relaxing, carbon dioxide (CO2) is expelled from the lungs.

The cardiovascular system is a "closed" system, which means that blood does not leave the vessels at any point. As noted in the preceding description of capillaries, diffusion allows oxygen and nutrients to pass through the vessel walls and into cells, and conversely, waste products and CO2 can pass from the cells through the vessel walls. The space in between blood vessels and cells is occupied by a fluid substance called *interstitial fluid* (i.e., *tissue fluid)* (302).

There are three types of blood cells: *red, white,* and *platelets*. Below are their primary functions:

- **Red:** carries oxygen to the body
- **White:** helps the body fight off infection
- **Platelets:** clots blood

Hemoglobin is a protein that is found in red blood cells. Its function is to transfer oxygen from the blood to the muscles. *Myoglobin* is a protein that is found in muscle that receives oxygen from the blood (red blood cells) and transports it to the mitochondria of the muscle cells (726).

Blood (i.e., red blood cells) goes to where it is needed the most. As a result, the legs are the main recipients of blood volume when running. At rest, 15–20 percent of one's total blood volume goes to muscles whereas during exercise, 85–90 percent of blood volume goes to muscles (727). The process by which this occurs is called *blood shunting*. This process dilates and constricts the arteries to determine what percentage of blood is received by working muscles.

HEART

This organ is the pump that supplies the body with blood and, therefore, oxygen. The heart is divided into four chambers (noted in fig. 4.1).

- **Right Atrium**
- **Right Ventricle**
- **Left Atrium**
- **Left Ventricle**

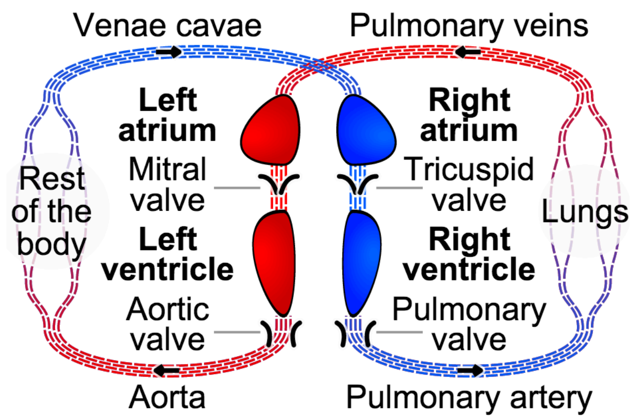


Figure 4.1 Circulatory system
http://commons.wikimedia.org/wiki/File%3ACirculatory_system_SMIL.svg

In figure 4.1, the four chambers of the heart are illustrated in the middle of the image (right and left atrium/ventricle). The blue lines denote deoxygenated blood (veins) and the red lines denote oxygenated blood (arteries). The circulatory system process:

1. The right atrium receives deoxygenated blood via the superior and inferior vena cava (veins).

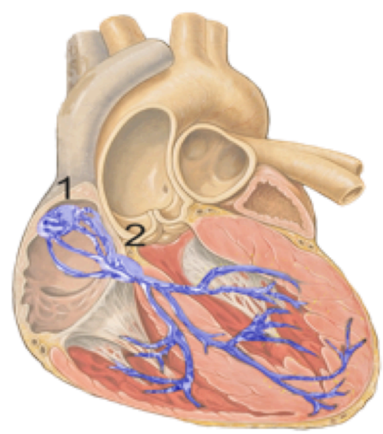
2. Blood passes from the right atrium through the tricuspid valve into the right ventricle, where it is pumped out to the lungs to be reoxygenated.

3. Reoxygenated blood returns from the lungs to the heart (left atrium).

4. Blood passes from the left atrium, through the mitral valve, and into the left ventricle.

5. Oxygenated blood in the left ventricle is pumped out the aortic valve and supplies the body with oxygen.

Heart contractions are caused by electrical impulses that originate at a spot on the heart called the **sinoatrial node (SA node).** This is the heart's natural pacemaker. The SA node sends an electric signal across the atria (i.e., left and right atriums) that causes a contraction. This electric signal also influences the *Purkinje fibers*, which stimulates the ventricles to contract. The Purkinje fibers are represented in figure 4.2 by the blue branches in the middle to lower parts of the heart (358).



1 – SA Node
2 – AV Node

Figure 4.2 Heart nodes

Image: Heart anterior view coronal section.jpg by Patrick J. Lynch (Patrick J. Lynch; illustrator; C. Carl Jaffe; MD; cardiologist Yale University Center for Advanced Instructional Media). Creative Commons Attribution 2.5 License 2007.

If the SA node fails, the **atrioventricular node (AV node)** takes over the pacemaking responsibility. Lastly, if the AV node fails, the Purkinje fibers can act as a weak pacemaker. The AV node primarily acts to regulate the signal from the SA node (357). The heart's rate of contraction is a function of the *autonomic nervous system* (ANS). This demonstrates how the heart has a built-in contingency system in case of failure of the SA or AV nodes.

Blood flow allocation within the body is based on need. For example, a runner would have greater blood flow to the legs than the arms because of the greater demand for oxygen in that region. This is most notable in individuals exposed to extreme cold for an extended period of time. In an effort to keep the body functioning, blood flow is concentrated to life-sustaining organs and diverted away from the extremities.

When exercising, heart rate is influenced by one's level of aerobic conditioning and the intensity of movement. The greater one's aerobic capacity, the more blood is ejected with each heart contraction, which translates to a lower heart rate compared with someone not as aerobically conditioned. An increase in heart rate correlates with an increase in physical activity because of the body's increased demand for oxygen.

Blood Pressure

Figure 4.3 Sphygmometer (blood pressure monitor)

A common measure of health, blood pressure represents the amount of pressure against the blood vessel walls. A blood pressure reading has two numbers, a top and bottom number. Normal blood pressure is 120/80. The top number is called the *systolic pressure* and represents the amount of pressure against vessel walls when the heart is contracting.

The bottom number is called the *diastolic pressure* and represents the pressure against the vessel walls when the heart is relaxing.

The numbers represent millimeters of mercury (mmHg), as this is the unit used to measure blood pressure. When taking one's blood pressure, the systolic number is the first beat that is heard and the diastolic is the last beat heard.

Under resting conditions, high blood pressure is considered 140/90 or higher. However, during exercise it is normal for blood pressure to increase substantially. Those with high blood pressure are often termed *hypertensive*. If your client is hypertensive, the person should be under a physician's care and guidance. Hypertension is typically managed by diet, exercise, and medication.

LUNGS

The lungs are located on either side of the heart. Deoxygenated blood is pumped from the heart to the lungs via the pulmonary artery. In the lungs, oxygen is diffused into the blood and carbon dioxide is transported out of the blood, where it is exhaled. This exchange of oxygen and carbon dioxide occurs in the alveoli, small air sacks in the lungs. Air (oxygen) is inhaled through the trachea (windpipe) and ends up in the alveoli, where it is used to reoxygenate the blood. The reoxygenated blood is returned to the heart via the pulmonary veins, where it is then pumped out of the heart and into the systemic circulatory system.

Contraction and relaxation of the diaphragm is the primary method by which the lungs inflate and deflate. As noted previously, when the diaphragm contracts, the diaphragm moves downward, allowing the lungs to fill with air. Conversely, relaxation of the diaphragm causes the air to flow out of the lungs (*think:* balloon deflating and losing its air).

Diaphragmatic Breathing

This type of breathing, also referred to as *belly* or *abdominal breathing*, is the most efficient way to breathe and relies on taking a full breath to make the diaphragm move downward and upward through its full range. Diaphragmatic breathing is characterized by the abdominal section of the body rising as the breath is inhaled. This is in contrast to *chest breathing* in which the lungs expand and make the chest rise.

The primary reason why diaphragmatic breathing is more beneficial over chest breathing (especially to athletes) is because more oxygen is inhaled and more carbon dioxide is exhaled with each breath. This equates to more oxygen to the working muscles (440).

❖ THREE ENERGY SYSTEMS

The energy system classification used by the body during exercise is determined by exercise intensity, duration, and fuel availability. The energy systems of the body largely function as a result of chemical reactions. For the purpose of this certification, it is not necessary to know all the specific chemical reactions that occur. However, it is important to know and understand the results of the chemical reactions on a macro level.

Understanding how the energy systems work in the body is of paramount importance to being a successful coach. By understanding these systems, you will be better able to design effective programs as it relates to intensity.

> While endurance sports focus largely on the aerobic (oxidative) aspect of training, anaerobic and neuromuscular aspects also contribute to the performance of an endurance athlete (589–591).

The body utilizes three classifications of energy systems during exercise:

- **Phosphagen**
- **Glycolytic**
- **Oxidative**

INTRODUCTION TO ENERGY SYSTEMS

At the most basic level, energy systems represent the body's ability to convert chemical energy stored in food into mechanical work. When discussing energy systems, we are really discussing how and in what manner ATP is produced. This is because in order for a muscle contraction to occur, ATP is required. The greater the amount of ATP that is supplied to working muscles, the longer the muscles can work. Conversely, the less ATP, the shorter the exercise session will be.

This relates to the two primary ways ATP is produced. If oxygen is not utilized, then ATP will be produced anaerobically, and if oxygen is utilized, then ATP will be produced aerobically. When ATP is produced anaerobically, it does not elicit a high number of ATP molecules for the working muscles to use as energy. However, when ATP is produced aerobically, many more ATP molecules are produced compared with the amount of ATP being produced anaerobically. This allows exercise to continue for a much longer period before fatigue occurs. Endurance sports such as distance running are primarily aerobic activities.

While ATP levels fluctuate based on the intensity of exercise, the levels of ATP never get to levels that would be considered dangerously low (561). While it is theorized that muscle is able to match ATP production and consumption, ATP levels are not stable during exercise, especially during high-intensity bouts (560).

Cells

The cells of the body take in raw material (nutrients from food) and convert them into energy. The macro view of how this process occurs is as follows (509):

1. Nutrients are broken down within cells
2. ATP is formed
3. When needed, ATP is broken down. This breakdown, or catabolism of ATP, creates usable energy for the cell.

Figure 4.4 illustrates how the body turns food into chemical energy (ATP).

The *Krebs cycle/oxidative phosphorylation* generates ATP aerobically, whereas the *Cori Cycle* generates ATP anaerobically.

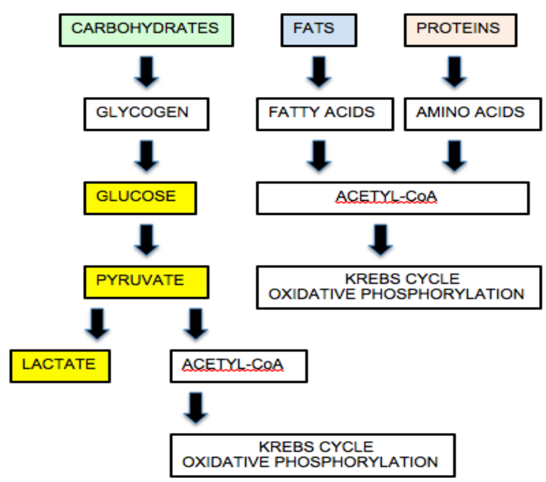


Figure 4.4 Energy system pathways

- The yellow boxes are representative of the components of the Cori cycle, discussed later.

While there are various theories regarding if the body can convert fat (fatty acids) to glucose, there is more substantial evidence to support that proteins can be converted to glucose via a process called *gluconeogenesis* (258). However, for unknown reasons, glucose that was converted from protein does not increase blood glucose levels (508).

FIRST ENERGY SYSTEM: PHOSPHAGEN

While all energy systems start at the same time, only one system typically provides the majority of the energy at any given time. Therefore when this certification denotes the first, second, and third energy systems, it is referring to the energy system performing the majority of work.

The first primary energy system is used for immediate energy, which is called the *ATP-CP system* or *phosphagen system*.

How Is Energy Generated?

Within a muscle cell exists small amounts of ATP as well as *creatine phosphate* (CP). Creatine phosphate is a high-energy compound that is stored in quantities five or six times greater than ATP (31). The creatine phosphate is broken down by an enzyme called *creatine kinese* (CK) that removes the phosphagen from CP and attaches it to ADP to form ATP (because of an added phosphate, “di” becomes “tri”) (31). This cycle repeats itself to provide the body with immediate, short-term energy. However, with each cycle, CP is reduced in the cell until ATP can no longer be resynthesized in this manner. At this point, the body moves to the second energy system (glycolytic).

Exercises that specifically utilize the phosphagen energy system are explosive and very short-term activities such as weightlifting or short, all-out sprints. These exercises are characterized by maximal but short duration efforts.

Since there is only a small amount of ATP within each cell, ATP must be resynthesized at the rate at which it is used.

SECOND ENERGY SYSTEM: GLYCOLYTIC

Below is a flow chart denoting the metabolism of carbohydrates. This is a detailed version of the carbohydrate energy pathway chart denoted above in figure 4.4.

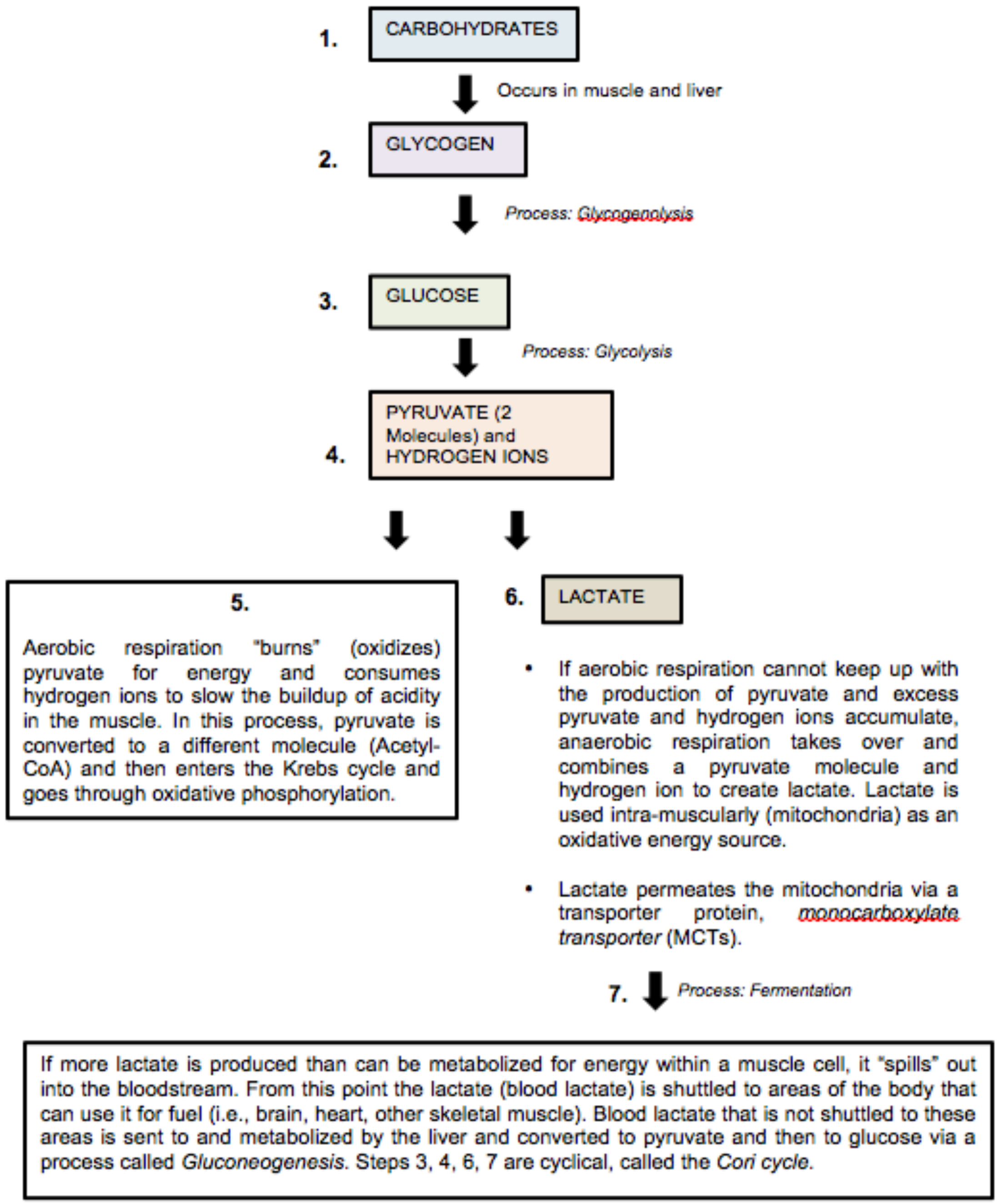


Figure 4.5 Carbohydrate metabolism (24, 29, 514)

This system uses lactate from the muscles and glucose from the blood and stored glycogen to provide energy to the muscles.

Glucose within the bloodstream (plasma glucose) can be used for immediate energy or can be stored in the liver and muscles in the form of glycogen. When the glycogen stores are called upon to provide energy, the glycogen is converted to glucose (29).

Plasma (blood) glucose typically stores 20 kcal of glucose or less. Compare this with liver glycogen stores, which range from 350–650* kcal, and leg muscle glycogen stores, which range from 1,250–2,270* kcal (based on 70kg body weight), and it is easy to see that plasma glucose, while technically a carbohydrate reservoir, is largely negligible (705).

* This range is based on normal glycogen levels and glycogen levels post-carbohydrate loading, respectively.

Glycogen in the liver can be distributed throughout the body where it is needed. Conversely, muscle glycogen can only be used within the muscle cell that it is being stored (705).

While anaerobic and aerobic/oxidative respiration occur simultaneously, the intensity of exercise largely determines the predominant energy pathway.

Blood Glucose Regulation

Like most aspects of physiology, the regulation of blood glucose (additionally termed: plasma glucose, blood sugar) is a series of checks and balances that looks to maintain homeostasis.

The primary players involved in maintaining blood glucose homeostasis are *insulin* and *glucagon*. Insulin is a hormone that is produced in the pancreas. When blood glucose levels get too high, insulin acts to reduce the concentration of blood glucose by facilitating the uptake of blood glucose into skeletal muscle and fat. As a result, fat is not used as an energy source.

Conversely, when blood glucose levels fall too low, glucagon, a hormone also produced in the pancreas, facilitates the increase in blood glucose level by converting glycogen in the liver to glucose, which is then released into the bloodstream. Blood glucose levels are also increased by ingesting carbohydrates.

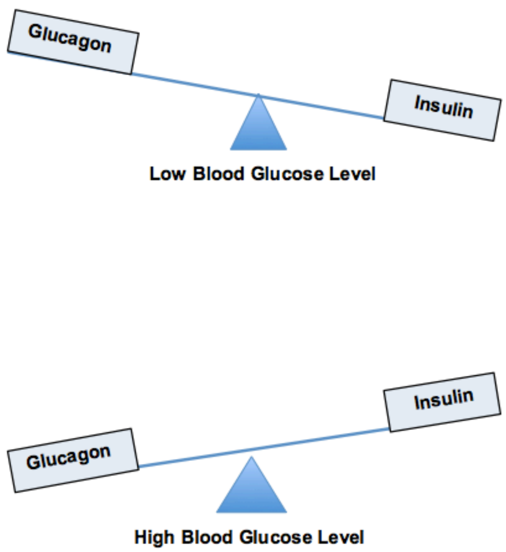


Figure 4.6 Glucagon/Insulin Response

IMPLICATIONS FOR RUNNING

Maintaining some degree of blood glucose/glycogen is required to not run out of energy (i.e., hit the wall). While blood glucose is not viewed as a substantial energy reservoir, studies have shown that ingesting carbohydrates while exercising can increase one's endurance by increasing the percentage of fuel contributed by blood glucose as opposed to stored glycogen (707, 708). Carbohydrates must be ingested well before one's suspected point of fatigue due to glycogen depletion (approximately 30 minutes). Interestingly, so long as the carbohydrates are ingested at least 30 minutes prior to fatigue setting in, the exact time when carbohydrates are ingested midrace has little influence in regard to their effectiveness (709).

Being in "good" aerobic condition correlates with the body's ability to more efficiently utilize fat as an energy source. This reduces an individual's reliance on carbohydrates as a fuel source, thus preserving more carbohydrates to use as fuel at a later time and/or at greater exercise intensities.

In regard to distance running events such as marathons, starting off the event with topped-off glycogen reserves is highly recommended to reduce the chance of hitting the wall. The process by which this occurs is *carbohydrate loading*. Topping off your glycogen stores prior to exercising is analogous to filling up your car's gas tank before a long trip. The theory behind carbohydrate loading as well as various loading methods is discussed in greater detail in the *Sports Nutrition* module.

Muscle Glycogen Storage

As noted previously, glycogen is primarily stored in the liver and the muscles. Between the two, a greater amount of glycogen is stored in the muscles versus the liver. While glycogen in the liver can be deployed throughout the body, glycogen in the muscles stays localized within the muscle. Therefore, specifically in regard to running, the larger the leg muscle mass (specifically the thigh), the greater the amount of glycogen that can be stored (705, 706).

While the liver size of individuals is typically the same (approximately 2.5 percent of total body mass) (714), the leg muscle mass of individuals varies substantially (18–22.5 percent in women, 14–27.5 percent in men) (711–713). As a result, the capacity of muscle glycogen storage between individuals will vary greatly based on their leg muscle mass.

Anaerobic Respiration

Often termed *fast glycolysis*, this process refers to steps 6 and 7 of figure 4.5. The formation of lactate (figure 4.5 – step 6) does not necessarily occur because of an absence of oxygen, but rather without a substantial utilization of it. This is an important distinction because the absence of oxygen is not a requirement for the formation of lactate (139).

1. Glycogen is converted into glucose via a process called *glycogenolysis* (33).
2. Through the process of glycolysis, glucose is converted into pyruvate.
3. If aerobic respiration cannot keep up with pyruvate production, lactate is formed from pyruvate and hydrogen ions.

4. Lactate is converted to pyruvate and then to glucose in the liver. The glucose is then shuttled back to the muscles via the blood to be used for energy.
5. The glucose in the muscles is converted back to pyruvate, then lactate.

Steps 3, 4, and 5 in the process are cyclical in nature and called the **Cori cycle** (29). This cycle is denoted in figure 4.7.

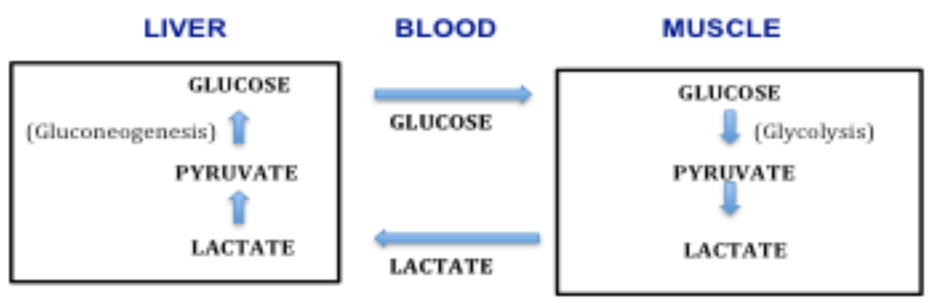


Figure 4.7 Cori cycle (515)

Aerobic Respiration

Often termed *slow glycolysis*, this process refers to step 5 of figure 4.5. In this process, pyruvate is converted into acetyl coenzyme A (acetyl Co-A) and then enters the Krebs cycle (34). This represents the beginning of the oxidative energy system. Acetyl Co-A is a molecule whose primary purpose is to facilitate energy production oxidatively.

THIRD ENERGY SYSTEM: OXIDATIVE

The third and final energy system is the *oxidative system*. This system has other names such as the *citric acid cycle*, *TCA cycle*, and *Krebs cycle*. This is the primary energy system utilized by distance runners and other endurance sport athletes.

The Krebs cycle is a series of chemical conversions of carbohydrates, fats, and proteins into carbon dioxide and water. In other words, the purpose of the Krebs cycle is to create an oxidative reaction. The resulting oxidative reactions from the Krebs cycle are then sent to the *electron transport chain* (ETC).

The Krebs cycle produces electron carriers. During the ETC, these carriers lose and gain electrons repeatedly as a way to transport energy. So what is the purpose of the ETC? The loss of energy by the carriers is used to create ATP from ADP via an enzyme called *ATP synthase*.

The ETC takes place in the membrane of mitochondria and is the last stage of the oxidative phase. It is responsible for the majority of ATP generation (34).

Converting Fat to Energy

An important aspect regarding the body's energy production is the use of fat for energy. The primary location for fat storage is **adipose tissue** (i.e., body fat) within the body. The majority of the adipose tissue used for energy is called *subcutaneous fat* (below the skin). This is not to be confused with visceral fat, which is fat located in the intra-abdominal cavity, also known as abdominal fat. This fat is what causes the protruding effect of some individuals' abdominals.

Adipose tissue is composed of **adipocytes,** or fat cells. Adipocytes are where *triglycerides* are stored and synthesized. Triglycerides are compounds containing glycerol and free fatty acids. Glycerol is a chemical compound that is the primary chemical support structure of triglycerides (37). Glycerol and free fatty acids are the elements *catabolized* (broken down) by the body for energy. Glycerol is broken down anaerobically through glycolysis, whereas fatty acids are catabolized through a process called **beta-oxidation**. During beta-oxidation, fatty acid is converted into *acetyl-CoA* and then enters the Krebs cycle to be oxidized.

Catabolism of fatty acid yields significantly more ATP than that of glucose (35).

Durations of Energy Systems

The following graph depicts the three energy systems and their contribution of total energy as it relates to duration of exertion. As noted previously, while all three systems start at the same time, typically only one predominant system is providing energy at any given time (39). For example, regardless of how intense an exercise level is, some amounts of ATP are generated aerobically (oxidative phosphorylation) (538). Therefore it is technically erroneous to say that the phosphagen system engages first followed by the glycolytic and then oxidative systems.

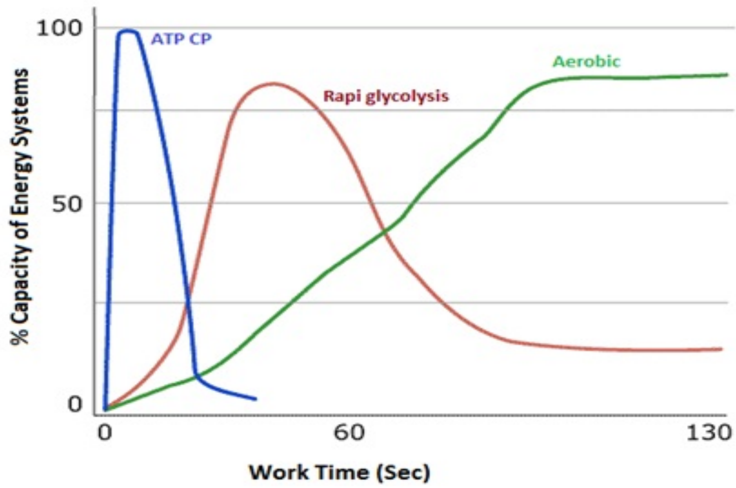


Figure 4.8 Energy system durations

Courtesy of Ikram Hamyd, www.ikramhyamyd.com

A specific example of how the different energy systems contribute to an exercise bout, regardless of the intensity, is the Wingate test (WAT). The WAT is a 30-second maximal effort cycling assessment designed to assess an individual's anaerobic fitness capacity. At least 20-30 percent of the energy utilized during the WAT comes from aerobic sources (558). This correlates with figure 4.8.

An analogy that describes how energy systems behave sequentially is a car with an automatic transmission. For the purpose of this analogy, the car has three gears.

- **First Gear (Phosphagen):** quick acceleration, but tops out quickly

- **Second Gear (Glycolytic):** medium-range gear; can be driven at high speeds but can only drive so far before it burns out

- **Third Gear (Oxidative):** long-range gear where the car is most efficient; slow to accelerate but is built to last for long hauls

Cars shift gears for the same reason energy systems do: efficiency and sustainability.

Glycolytic and Oxidative Relationship

While the aforementioned automatic transmission analogy identifies glycolytic and oxidative energy systems as two separately occurring phases, they occur simultaneously as is noted in the following chart. While the majority of the energy source is based on the intensity of the exercise, even at low intensities, glycogen is still an energy source.

As much more ATP is generated from fat than from carbohydrates, it would be great if fat could be the sole energy source and save the carbohydrate (glycogen) stores for when they are needed. However, glycogen in varying amounts is always being utilized for energy (705).

The following chart (figure 4.9) demonstrates the fuel source (fat, carbohydrate) based on the intensity of exercise in relation to one's VO_2 max. This illustrates that at all aerobic exercise intensity levels, both fat and carbohydrates are energy sources, in varying percentages based on the intensity level (705).

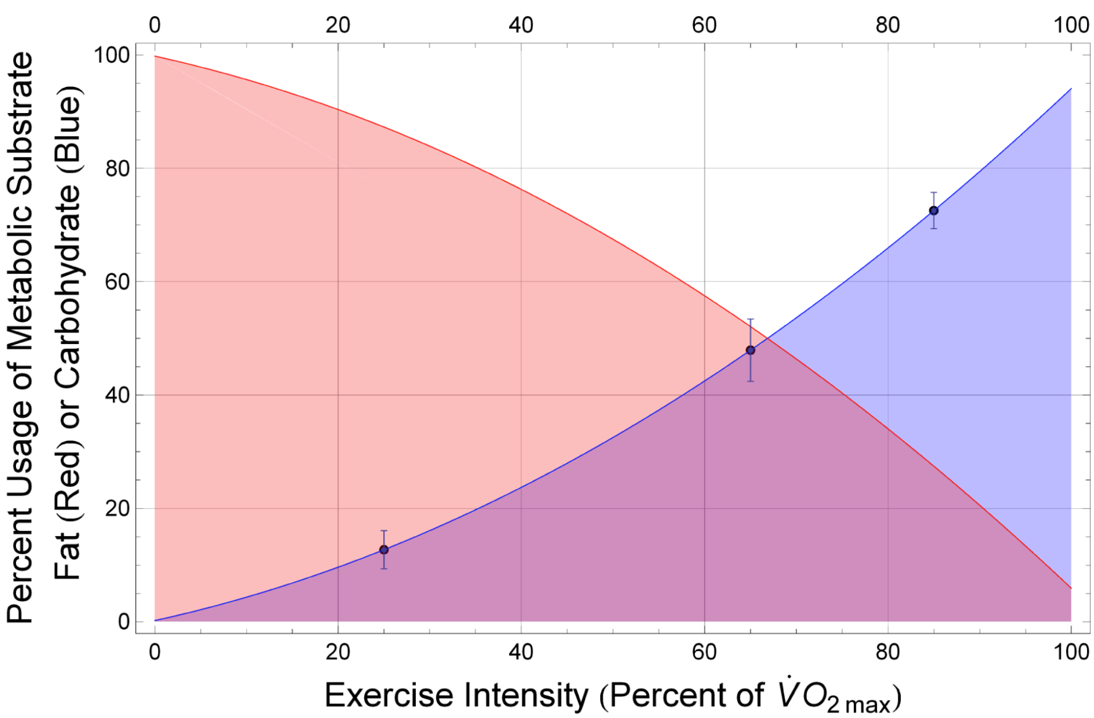


Figure 4.9 Fat/Carbohydrate Fuel Source

PHYSIOLOGICAL RESPONSE TO AEROBIC TRAINING

As a running coach, it is important to understand the physiological effects that aerobic training has on an individual. Below are physiological adaptations that occur as a result of aerobic training (36):

- Decrease in resting heart rate
- Increase in stroke volume
- Greater capacity to use fat as fuel
- Less reliance upon blood glucose for fuel
- Increase in capillary density
- Increase in mitochondrial density

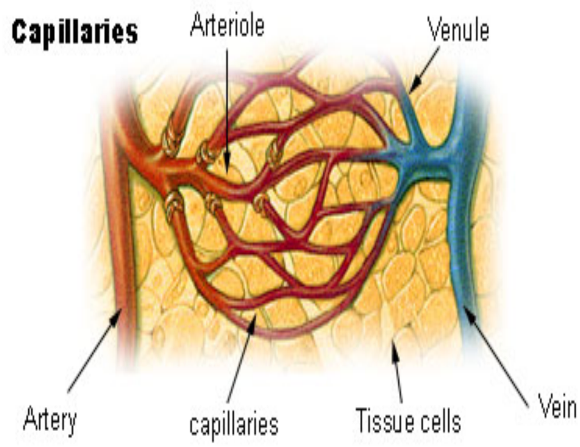


Figure 4.10 Capillary Bed

http://training.seer.cancer.gov/module_anatomy/images/illu_capillary.jpg

The first two adaptations are cyclical in nature, as an increase in stroke volume results in a decrease in resting heart rate. An increase in stroke volume means that the heart is pumping more blood per stroke, thereby increasing its efficiency. Because more blood is being pumped per stroke, the absolute number of strokes is reduced, hence, lowering the resting heart rate.

Capillaries are able to increase in number and size in areas where additional blood supply (i.e., oxygen) is needed. Because of the oxygen requirements of endurance athletes, well-trained athletes have a much greater capillary density than nonactive individuals. Capillaries grouped together are called a **capillary bed** (38).

The greater the number of capillaries, the larger and denser the capillary bed is. As the capillary bed becomes larger, the greater the oxygen supply to muscles becomes because of the increased blood flow. As noted previously, a runner would have much greater capillary density in the legs versus the arms. For most runners, the arms would have similar capillary density to a nonendurance athlete.

Another adaptation of aerobic training takes place on a cellular level and is represented by an increase in *mitochondrial density*. **Mitochondria** are the structures within cells that produce **adenosine triphosphate (**ATP). Well-trained skeletal muscle (such as in runners) has an increased capacity to generate ATP aerobically relative to nontrained skeletal muscle because of increased mitochondria.

Signs of Being Aerobically Deconditioned

Deconditioned individuals typically present with some or all of the following physiological effects, as compared with those who are aerobically conditioned:

- Higher heart rate at rest and while exercising at low to moderate intensities
    - Until VT/LT is reached, one's RPE is often low in relation to the heart rate. Once VT/LT is reached the RPE typically elevates rapidly.
    - Heart rate at moderate intensity is at a similar level to someone who is aerobically conditioned at a high intensity.
- Slow heart rate recovery

HYPOXIC AND HYPEROXIC TRAINING

The oxygen saturation levels at which your clients train has profound physiological effects on their bodies, as well as their performance capabilities.

Hypoxic Training: Training that occurs in an environment that is substantially depleted of oxygen (e.g., training at high altitude). It can be done in either natural (altitude training) or simulated environments (altitude tent).

Hyperoxic Training: The opposite of hypoxic training, meaning that the training is done in the presence of a dense oxygen supply. This can be done only in simulated environments using supplemental oxygen.

Hypoxic Training

While altitude training could be considered a training tool, given the context in which we will be discussing it, we will include it in the physiology section. Altitude training first became a hot topic after the 1968 Olympics in Mexico City, Mexico, where most endurance events had subpar performances and most sprint and other anaerobic events had record-breaking times (50, 52).

Mexico City sits at an elevation of 7,350 feet above sea level (450). High-altitude training is defined as training at altitudes of 5,000 feet or above. What was discovered when researching the 1968 Games was that because of the lower concentration of oxygen at 7,368 feet, the athletes did not have as much oxygen to breathe, which limited their performance in endurance events. While participants of anaerobic events (e.g., sprints) were subjected to the same diminished oxygen levels, it did not impact them as much since their reliance on oxygen was minimal given the anaerobic nature of their event(s).

It was determined that the reason why anaerobic athletes broke so many records was because the air was less dense at the higher altitude compared with sea level, and therefore, there was less air resistance, which led to faster times. The 1968 Games were the catalyst in determining what physiologically occurs in the body at altitude and how it can be exploited to enhance human performance (50).

When an individual is at altitude, the primary physiological effect that occurs is a decrease in oxygen hemoglobin saturation (49). *Hemoglobin* is what transports oxygen in the blood from the lungs to the muscles of the body. The body's response to this decrease in hemoglobin is to increase the production of *erthyropoietin* (EPO), a hormone secreted by the kidneys that stimulates red blood cell production (51). This is the primary reason why athletes train at altitude. The thought process is that by training at altitude, athletes will increase their red blood cell count and therefore have capacity to deliver more oxygen to working muscles. This increase in oxygen to muscles enables the athlete to perform at a higher level.

The effects of altitude training have been shown to last for up to 15 days after athletes return to sea level (48). Therefore, many athletes schedule their altitude training so that they still have the benefits of the altitude training when competing at lower altitude. It is important to note that the secretion of EPO can be increased with altitude training, but only by a finite amount. If too much EPO were to be secreted by the kidneys, too many red blood cells would be produced and this could have harmful side effects. This is an example of how the body regulates itself.

The term *hematocrit* refers to the volume (percent) of red blood cells in blood. Through hypoxic training, the goal is to raise one's hematocrit level, thereby increasing the amount of oxygen delivered to working muscles. A male's hematocrit level is approximately 45 percent, whereas a female's is typically 40 percent (381).

There is no refuting that altitude training stimulates red blood cell production, but because of decreased oxygen levels, an individual cannot train as hard in that environment. Therefore, the question is if the benefits of increased red blood cell production are canceled out while training at altitude. Because of this a new theory has emerged. This theory is termed *Sleep High, Train Low* (53). Some of you might be thinking that this seems a bit odd and find yourself wondering, *How can you sleep at altitude, then train at sea level?* You are right that this is highly impractical.

The logic behind this concept is that by sleeping at altitude you will gain all the benefits of high-altitude training (increase in red blood cells) but will be able to train at sea level, where you can train at a normal intensity. This type of training is possible only by using altitude tents, which decrease the oxygen level that you breathe in. This practice has proved to provide the best of both worlds. Some of these tents can replicate altitude as high as 12K feet (275).

Altitude tents are also helpful if an individual is heading to high altitude and needs to acclimate prior to the trip. If clients are planning on utilizing an altitude tent or going to altitude to train, it is advised for them to consult with their doctor prior to performing this type of training.

It is important to note that individuals will adapt to altitude at different rates.

Training in the heat may also simulate training at altitude. Because of the body's need to cool itself, blood is shunted away from the working muscles and toward the skin. This reduction in blood flow equates to less oxygen to the working muscles during exercise – similar to the effects of training at altitude. This is one reason why exercising in the heat is more difficult than at lower temperatures. The physiological adaptation that occurs from training in the heat is an increase in blood plasma volume (580, 581). An increase in blood plasma volume may increase stroke volume and thus lower exercise heart rates (582).

Blood Doping

This certification mentions doping because it is an illegal method by which to increase one's red blood cell count, thereby enhancing the body's ability to get as much oxygen to the muscles as possible. Blood doping is not only illegal but also extremely dangerous.

One type of doping is done via blood transfusions. Blood transfusions attempt to accomplish the same goals as altitude training, but in much less time. There are two primary types of blood doping. The first is transfusing blood – blood removed from an individual and injected into an athlete at a later date. The blood can be from either a donor (*homologous*) or that athlete (*autologous*). The risks are extremely high with this type of blood doping, but the desired effects are the same as those of altitude training: to increase the number of red blood cells.

A second type of doping is accomplished by using *synthetic erythropoietin* (EPO) to boost red blood cell production. As mentioned previously, EPO is a hormone that occurs naturally in the body and is secreted in higher levels by the kidneys during altitude training. The primary risk of using EPO and transfusing blood is the possibility of *polycythemia* – abnormally high levels of red blood cells (RBC) (387). High levels of RBCs increase the viscosity of blood and can cause *hypertension* (high blood pressure), which can lead to a stroke or heart attack.

Effect of Supplemental Oxygen

If a decrease in oxygen is detrimental to aerobic performance, then the opposite must be true. This is called *hyperoxic training*. In a study done at the University of New Mexico, five experienced cyclists were asked to train at 1,600 meters (one mile above sea level) at varying intensities until no improvement in endurance was noted and they had reached a physiological plateau (54). Depending on the athlete, it took between seven and 34 weeks to reach this plateau. To attempt to break this plateau, the subjects were asked to perform intervals at 95 percent of their maximum heart rate. This attempt, however, was unsuccessful as none of the subjects was able to sustain the training sessions for the required amount of time. At this point, the subjects were given 70 percent supplemental oxygen (the air we breathe is 21 percent oxygen).

The result of the hyperoxic training was very encouraging. Not only were the subjects able to complete the 95 percent intervals, after hyperoxic training for six weeks, the subjects were able to increase their endurance by 32 percent while riding from 90–95 percent of their maximum heart rates. Furthermore, during high-intensity pedaling, the subjects' heart rate decreased by an average of five beats per minute (54). As all of the subjects had reached a plateau prior to the hyperoxic training, it is probable that the hyperoxic training was responsible for the improvement in their performance at high altitude.

It should be noted that hyperoxic training is not a standard training practice. Regardless, in order for hyperoxic training to occur, the following criteria must be met:

1. Individual administering oxygen is certified to do so
2. Done under clearance from physician
3. Must be deemed legal by local/state laws and sport governing body

❖ EFFECTS OF INTENSE EXERCISE

Runners who strenuously exert themselves for long periods run the risk of encountering one or all three of the following issues: *glycogen depletion*, *oxygen debt,* becoming *"out of breath."* The common theme among these three things is that they are all the result of intense exercise. Additionally, all of these things relate to being fatigued. *Muscle fatigue* is a general term that in regard to running equates to a reduction in the force that is generated by a runner.

GLYCOGEN DEPLETION

Glycogen depletion is a technical term for *hitting the wall*, *bonking*, or *blowing up* … get the picture? If you are an endurance athlete, you have probably experienced glycogen depletion at some point, and if so, you also know why it is termed *hitting the wall*. This is because glycogen depletion feels like you ran into a wall and are stopped dead in your tracks.

Glycogen depletion occurs when the muscles and liver are empty of all glycogen and the body must rely solely on fat for energy. While individuals can continue, they must do so at a much lower rate of intensity. The reason for this is that converting fat to use as an energy source requires a substantial amount of oxygen and thus occurs at a relative low intensity level.

A 2010 study by Rapoport noted that many exercise physiology resources infer that because of insufficient glycogen stores, hitting the wall during a marathon is inevitable unless a runner's glycogen stores are full because of carbohydrate loading (705). However, what Rapoport found was that there are many factors that play into one's ability to complete a marathon in regard to glycogen reserves, such as:

1. Muscle mass distribution
2. Liver and muscle glycogen density
3. Running speed

While even the leanest of athletes have enough nonessential fat to run multiple back-to-back marathons, the primary performance limiter is carbohydrates (glycogen) and, more specifically, their relative small reserves compared with fat (705). This is an issue as distance running utilizes substantial carbohydrate reserves.

Glycogen Depletion and Marathons: The Statistics

Approximately two in five runners report hitting the wall during a marathon and 1-2 percent drop out before finishing (705).

Generally speaking, the more aerobically conditioned runners are, the more efficient they are at utilizing fat instead of glycogen for energy. Conversely the less aerobically conditioned athletes are, the more reliant they are on glycogen. The average male has about 90,000–110,000 kcal of energy from fat reserves (1 gram of fat has 9 kcal of energy), whereas the carbohydrate energy (glycogen) reserve is less than 2,000 kcal in a non-carbohydrate-loaded state (302). This clearly demonstrates that untrained individuals who rely on glycogen as an energy source will find that they cannot exercise for very long at a medium to high intensity before running out of energy. As an analogy to this, think of burning a piece of paper versus a wood log. The piece of paper catches fire immediately but burns out quickly. The log, however, burns for quite a while. In this analogy the piece of paper is glycogen and the wood is fat.

As mentioned earlier, the two primary locations where glycogen is stored are the liver and muscles. While the concentration of glycogen in the liver is higher than in muscles, the absolute amount of glycogen stored in the muscles is 75 percent versus 25 percent in the liver (41). If glycogen were utilized as the sole energy source, an athlete would be able to run for approximately two hours before total glycogen depletion occurs, depending on the intensity level. However, if a runner started off with low glycogen levels, the time to depletion would be less.

The ability to utilize fat as the primary source of energy instead of glycogen is important because an endurance athlete needs an energy reserve. This reserve is glycogen, and if it is used up due an athlete's inefficiency to use fat as fuel, the performance of the athlete will decrease substantially.

As glycogen depletion requires a runner to slow down or stop, from a performance standpoint this is often viewed as catastrophic. As a result, proper pacing over the course of a long event, such as a marathon, is critical to minimizing the chance of *hitting the wall*.

Many factors are involved in human performance as it pertains to not ending up in a glycogen-depleted state. Some of these factors are nutrition (carbohydrate loading and prerace fueling), pacing, and other related risk factors. These areas are discussed in the *Sports Nutrition* and *Pacing* modules, respectively.

Glycogen-Depletion Training

Low glycogen is typically viewed as a bad thing in respect to endurance sports, as it represents running low on energy and moving closer to the dreaded *bonk*. However, there is a training practice called **glycogen depletion training (GDT)** that has individuals perform physical training sessions while in a glycogen-depleted state (no glycogen before or during activity).

The purported benefit of GDT is to reduce the body's reliance on glycogen by making it more efficient at utilizing fat as a fuel source.

While there is not a lot of research in this area, a study by Van Proeyen et al. found that individuals who performed GDT had increased fat utilization over a group that ingested a carbohydrate-rich breakfast. Additionally, the GDT group had increased enzymes associated with fat metabolism while the non-GDT group did not (643). It should be noted that the non-GDT group did not show increased performance over the GDT group.

THE SCIENCE BEHIND GDT

The stress of GDT increases the activity and amount of a protein (PGC-1α) that is responsible for increasing fat oxidation and production of mitochondria (645).

Training in a glycogen-depleted state stresses the body. As a result, GDT should not be undertaken on a regular basis as it is likely that the repeated stress will decrease overall energy and performance. GDT is to be used sparingly and solely during training, not in competition.

Lastly, as the purported benefit of GDT is to increase the body's efficiency of utilizing oxygen as an energy source, GDT is more applicable to long-distance events such as half and full marathons. For shorter events such as 10K and less, it can be theorized that GDT would not have a substantial benefit, as glycogen plays a larger role in providing energy for these types of events (644).

GDT is most often implemented by athletes who have a history of bonking late in a race.

It is advised that when performing GDT, athletes run on a treadmill or on a course that is relatively short so that if and when they bonk, they are not too far from home or are close to a source (i.e., restaurant) where they can refuel. Carrying emergency glycogen such as a gel or carbohydrate drink is strongly recommended.

OXYGEN DEBT

Anyone who has ever exercised at a high degree of intensity has experienced oxygen debt. Oxygen debt is most often characterized by heavy breathing after an intense exercise bout ends.

The definition of oxygen debt is the difference between oxygen consumption at rest and elevated rate of oxygen consumption following an exercise bout. When an exercise session ends, the level of oxygen consumption does not immediately return to a resting baseline. There is a slow return to this baseline level. Oxygen debt is indicative of this return to a baseline.

Perhaps the best way to think of oxygen debt is in terms of a credit card. If you use your credit card to purchase an item, you have accrued debt and owe this amount to the credit card company. Once you have paid off the credit card in full you have a balance of zero. This is analogous to oxygen debt because one must repay the oxygen used until the body returns to a zero baseline.

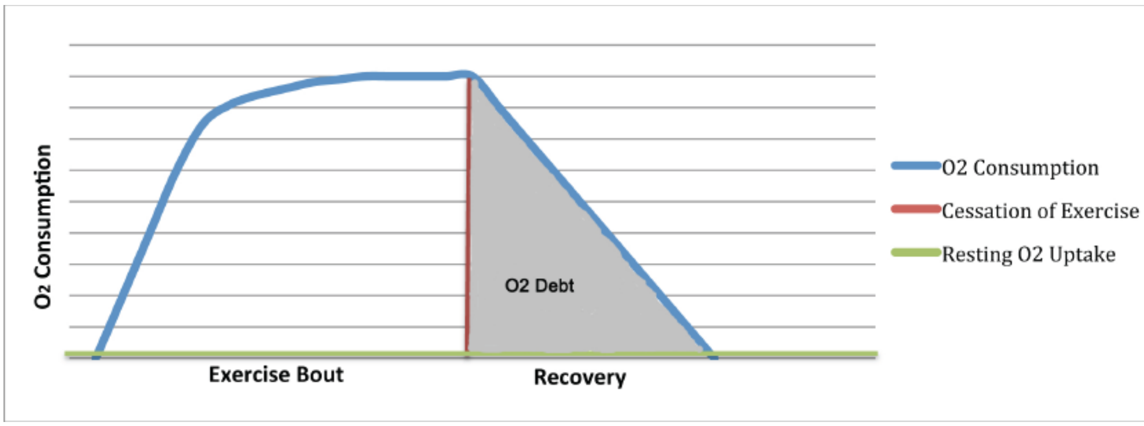


Figure 4.11 Oxygen debt

The repayment of oxygen is termed *excess post-exercise oxygen consumption*, commonly referred to as EPOC. EPOC represents the intake of oxygen after exercise ceases and the purpose of EPOC is to reduce and eliminate the oxygen debt, thus returning the body to normal (nonexercising) oxygen levels. EPOC occurs with both aerobic and anaerobic exercise. The higher the intensity level of exercise is, the greater the oxygen debt and EPOC (42).

> The reason why high-intensity training that is primarily anaerobic also trains the aerobic energy system is due to EPOC. The oxidative energy system (aerobic) works very hard to bring an individual back to baseline after intense anaerobic work. This is primarily why high-intensity, interval-based training elicits increases in both anaerobic and aerobic fitness levels.

Studies have shown that while the oxygen level consumed at rest (i.e., the baseline) is similar between trained and untrained individuals, aerobically trained individuals return to baseline faster than untrained individuals. This is because untrained individuals rely more on anaerobic energy systems during strenuous exercise than trained individuals, and therefore the resulting oxygen debt is higher.

Three areas that oxygen debt addresses when "repaying" debt are (120):

1. Oxidizing lactate
2. Supplying the demands of an increased metabolic rate
3. Replenishing high-energy phosphate stores
    a. Refer to *First Energy System: Phosphagen*

'OUT OF BREATH'

From a postural standpoint, positions can decrease or increase the lungs' ability to get a full breath. However, it is interesting to note that studies have shown there is little difference between sedentary and trained individuals' pulmonary function (43). Two areas of pulmonary function are *total lung capacity* (TLC) and *forced vital capacity* (FVC). TLC represents the largest volume of air the lungs can hold, and the FVC represents the largest amount of air that can be expelled after taking a deep breath. So, if pulmonary function is not substantially increased with training, why do people who are deconditioned become "out of breath" faster than those who are aerobically well trained?

The major limiting factor in exercise is cardiac output (44). Cardiac output pertains to the volume of blood being pumped out of the heart in one minute. There are two ways that cardiac output can be increased:

1. Increase in heart rate
2. Increase in stroke volume (the amount of blood being pumped out of the heart with each stroke)

Aerobically conditioned individuals have a higher cardiac output than aerobically deconditioned individuals, and are therefore able to supply more oxygen to their working muscles than those who are aerobically deconditioned. Therefore the primary reason why a deconditioned individual gets out of breath faster than someone who is aerobically conditioned is because his or her body does not utilize oxygen for energy efficiently (43).

Being out of breath usually coincides with increased and relatively shallow breathing. Untrained individuals who exercise with intensity often do not have optimal oxygen delivery to the muscles because of inadequate cardiac output. The body naturally looks to compensate for this lack of oxygen by increasing the number of breaths that are taken in an attempt to increase the oxygen saturation of the blood. This increase in respiration rate is what most people relate to being *out of breath*. The point where a person becomes out of breath is also called the *ventilatory threshold* (VT). It is important to note that anyone who exercises at high intensity can experience being out of breath, regardless of his or her conditioning level. However, aerobically conditioned individuals will reach this level much later than those who are aerobically deconditioned.

❖ SUMMARY

- The body produces substantially more ATP aerobically than anaerobically.
- The *Cori cycle* is the process by which the body produces ATP anaerobically.
- The three energy systems are *phosphagen*, *glycolytic,* and *oxidative.*
- The body stores only a small amount of ATP at any given time. Therefore, ATP must be continually regenerated on a cellular level.
- Aerobic training results in both increased mitochondrial and capillary density.
- Fat yields more ATP generation than glucose.
- *Lactate threshold* (LT) is representative of the level at which blood lactate accumulates in the bloodstream. This occurs when lactate production exceeds lactate clearing.
- Therefore other measures (i.e., lactate threshold) are often used to infer AnT.
- The two main circulatory systems in the body are:
    - Pulmonary
    - Systemic
- Red blood cells carry oxygen to the body.
- The *sinoatrial node* (SA node) is the body's natural pacemaker.
- Diaphragmatic breathing is beneficial over chest breathing.
- Insulin and glucagon work together to maintain homeostasis.
- Glycogen depletion occurs when the muscles and liver are empty of all glycogen.
- The definition of oxygen debt is the difference between oxygen consumption at rest and elevated rate of oxygen consumption after an exercise bout.
- *Hypoxic training* is done to increase one's red blood cells. This results in the body being able to deliver more oxygen to working muscles.
- An aerobically trained individual will reduce oxygen debt faster than someone who is aerobically deconditioned.
- *Cardiac output* is the primary factor in an individual's being "out of breath," not pulmonary function.
- Aerobically deconditioned individuals often have elevated heart rates and slow heart rate recovery times.
- An anaerobic state does not mean a lack of oxygen but rather the lack of oxygen utilization.